DP Biology – A3.2 (HL only)

# Classification and cladistics

- Which tools help us sort organisms into taxonomic groups?
- How does cladistics differ from traditional classification?

HL statements A3.2.1–9. Textbook pp. 125–137.
Image: a stoat. Steve Hillebrand, USFWS, public domain.



# Objectives for each class

| Class | Statements | By the end of the class, you can… |
|---|---|---|
| **1** | A3.2.1–3 | Explain why we classify, the problems with ranked taxa, and the value of classifying by ancestry |
| **2** | A3.2.4–5 | Explain clades, and how sequence differences act as a molecular clock |
| **3** | A3.2.6–7 | Build a cladogram from sequence data and analyse cladograms |
| **4** | A3.2.8–9 | Use cladistics to test classifications, and explain the three domains |

# Why classify?

A3.2.1



Arum maculatum: one species, dozens of common names. H. Zell, CC BY-SA 3.0.

- There are millions of species: grouping them makes the diversity manageable
- Each species has one **binomial** name, the same in every language
- Classification helps us identify organisms and find information about them
- It lets us **predict** features of a newly found species from its group

### Try it

This plant is called lords-and-ladies, cuckoo-pint and many other names. Why is one scientific name better?

# The traditional hierarchy of taxa

A3.2.2

| Rank | Grey wolf | Common daisy |
|---|---|---|
| **Kingdom** | Animalia | Plantae |
| **Phylum** | Chordata | Angiospermophyta |
| **Class** | Mammalia | Magnoliopsida |
| **Order** | Carnivora | Asterales |
| **Family** | Canidae | Asteraceae |
| **Genus** | Canis | Bellis |
| **Species** | Canis lupus | Bellis perennis |

- Seven main ranks, with domains above
- Each rank is nested inside the one above
- Genus and species give the binomial name

# Problems with ranked taxa

A3.2.2

- Evolution produces a branching tree, not neat levels: there is no natural rule for what counts as a family or an order
- Ranks are not equal: one genus may hold one species, another hundreds
- Similar features can evolve separately (**convergent evolution**), so groups based on appearance can mix unrelated organisms
- Traditional groups such as reptiles leave out descendants (birds) of their common ancestor
- Hybrids and species that are still diverging do not fit clear boundaries

**Key idea**

The traditional hierarchy does not always match the pattern of divergence that evolution actually produced.

# Classifying by evolutionary relationships

A3.2.3

- Every member of a group has evolved from one **common ancestor**
- So the group shares features inherited from that ancestor
- We can **predict** features of a member that has not been studied
- Classification shows evolutionary history, not just appearance

### Example

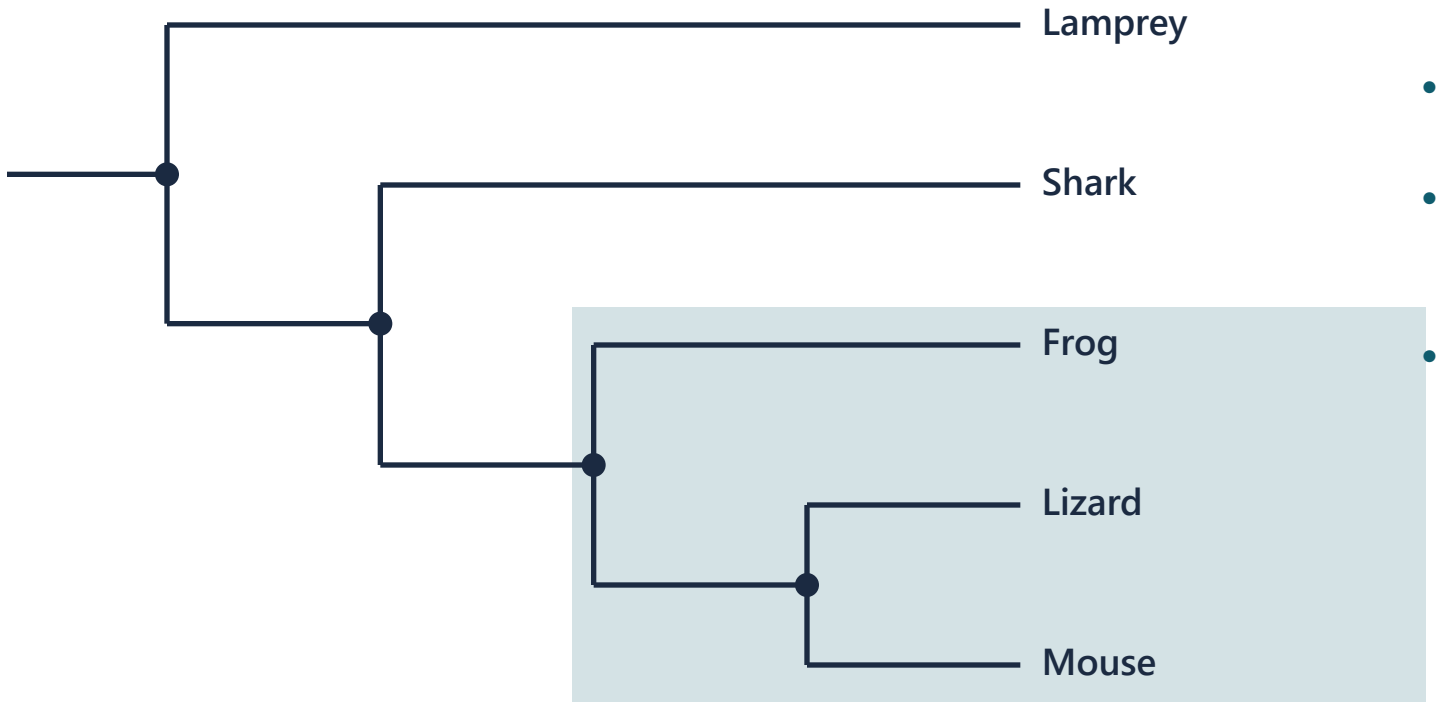
Bats fly like birds, but share hair, milk and a three-bone middle ear with other mammals. Grouping them with mammals lets us predict that a new bat species will also produce milk.

### Link

Evidence for evolution and the origin of species (A4.1).

# Clades

A3.2.4




Shaded: one clade, a common ancestor and all its descendants.

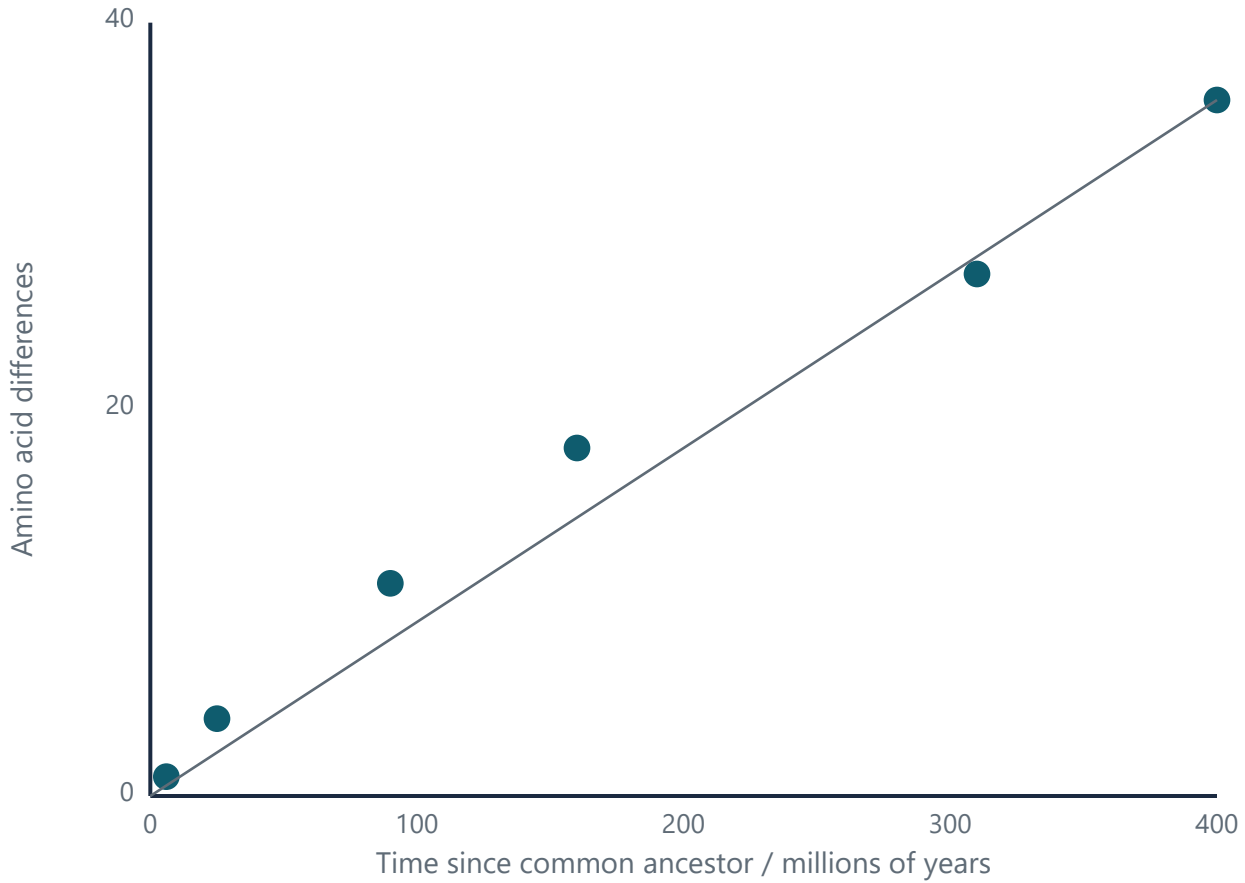
- A **clade** is a group made of one common ancestor and **all** of its descendants
- Members share characteristics inherited from that ancestor
- The most objective evidence comes from **base sequences of genes** and **amino acid sequences of proteins**
- Morphology can also be used, with care

**Try it**

Is "Shark, Frog and Lizard" a clade? Explain using the diagram.

# The molecular clock

A3.2.5




Schematic: differences in one protein between pairs of species.

- Mutations build up in a gene at a roughly steady rate
- So the number of differences between two species estimates how long ago they diverged
- Calibrated with fossils of known age

### Only an estimate

Mutation rates vary with generation time, population size, strength of selection and the gene studied.

# Building a cladogram from sequences

A3.2.6

| Species | Base sequence (12 bases) |
|---|---|
| W | ATGCCTAGGTCA |
| X | ATGCCTAGCTCA |
| Y | ATGACTAGCTGA |
| Z | TTGACAAGCAGA |

| | W | X | Y | Z |
|---|---|---|---|---|
| W | – | 1 | 3 | 6 |
| X | | – | 2 | 5 |
| Y | | | – | 3 |

Number of differences between each pair.

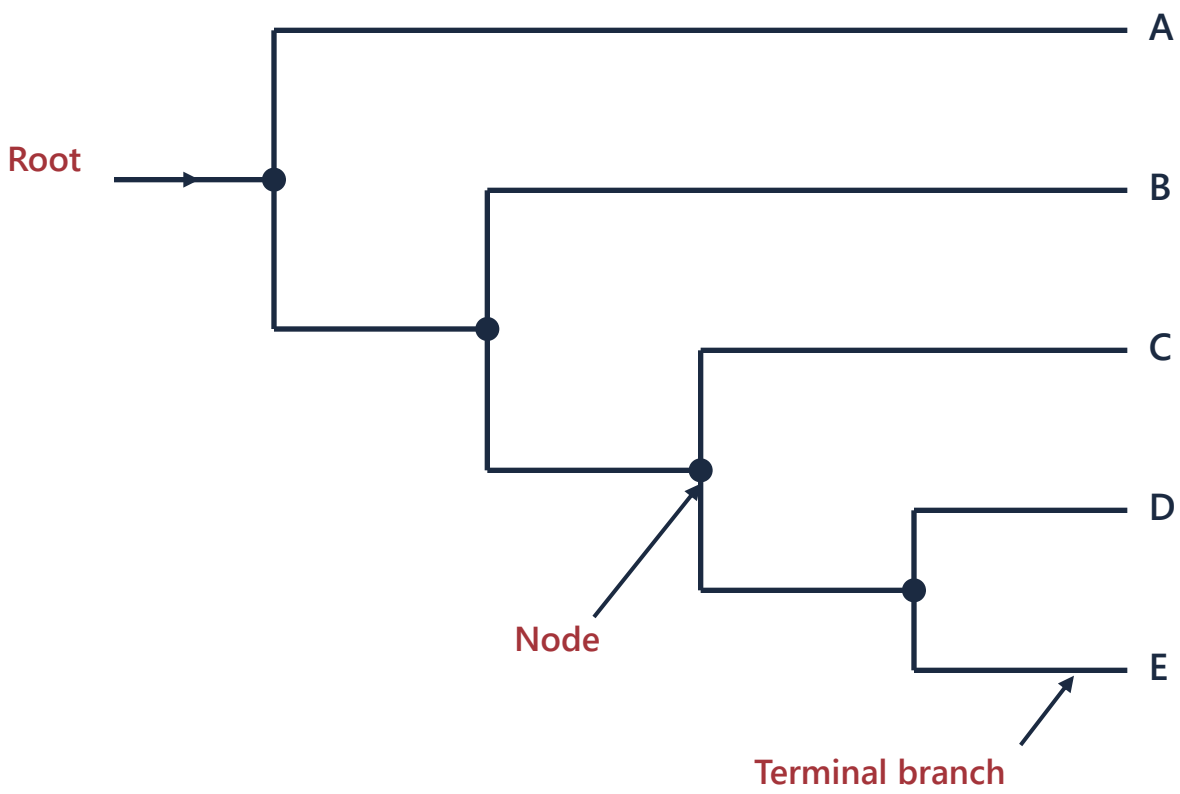
- 1. Count the differences between every pair
- 2. Join the two most similar first (W and X), then add the next closest (Y), then the least similar (Z)
- Fewest differences = most recent common ancestor

**Try it**

Draw the cladogram for W, X, Y and Z.

# Reading a cladogram

A3.2.7




- **Root**: the base, the common ancestor of all
- **Node**: a branch point, a hypothetical common ancestor
- **Terminal branch**: leads to one taxon
- Species sharing a more recent node are more closely related
- D and E are more closely related to each other than either is to C

**Try it**

Which taxa form the smallest clade that includes B and D? How many clades include C?

# Testing a classification: the figworts

A3.2.8



Figwort, Scrophularia nodosa. H. Zell, CC BY-SA 3.0.



Greater plantain, Plantago major. Public domain.

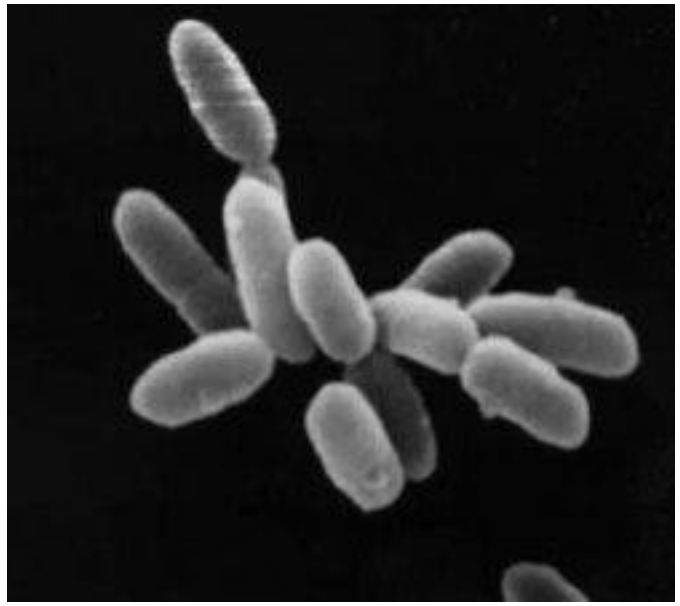
- The figwort family was based on flower shape
- DNA sequences of several genes were compared
- The family was not a clade
- Many genera, such as foxgloves and speedwells, moved to the plantain family

## Key idea

Cladistics can show that a traditional group is not a single clade. The classification is then changed so that groups match evolutionary relationships. The figwort family lost over half its species.

# Three domains

A3.2.9



Halobacterium, an archaean from salty water. NASA, public domain

| | Bacteria | Archaea | Eukarya |
|---|---|---|---|
| **Nucleus** | No | No | Yes |
| **Histones on DNA** | No | Some | Yes |
| **Introns** | Rare | Some | Common |
| **Membrane lipids** | Ester-linked | Ether-linked | Ester-linked |

## The evidence

In 1977, comparisons of ribosomal RNA base sequences showed that prokaryotes form two groups as different from each other as either is from eukaryotes. A new rank, the domain, was added above kingdoms: Bacteria, Archaea and Eukarya.

# Check-in: classification and cladistics

A3.2.1–9

1. List the main taxa in order from domain to species.
2. Explain one problem with ranked taxa.
3. Define a clade.
4. Explain how a molecular clock works and one assumption it makes.
5. Explain how to read relationships from a cladogram.
6. State the three domains and one difference between archaea and bacteria.

**How to use this**

Answer without your notes, then compare with a partner.

# Working with data: building a cladogram

A3.2.8

Number of base differences in a short DNA sequence between four species. These data are invented for practice.

| | W | X | Y |
|---|---|---|---|
| **X** | 2 | | |
| **Y** | 7 | 6 | |
| **Z** | 12 | 11 | 10 |

1. Identify the two most closely related species.
2. Draw a cladogram for W, X, Y and Z.
3. Explain why parsimony is used to choose between cladograms.

# Common mistakes to avoid

| Not this | But this |
| --- | --- |
| **Species at the tips evolved from each other.** | They share common ancestors at the nodes. |
| **Organisms on the left of a cladogram are more primitive.** | Order along the tips means nothing; branch points matter. |
| **Fungi are in the plant kingdom.** | Fungi are a separate kingdom, closer to animals. |

# Check that you can...

1. **Outline** the hierarchy of taxa and its problems
2. **Explain** classification based on evolutionary relationships
3. **Define** a clade and **outline** the molecular clock
4. **Construct** and **interpret** cladograms
5. **Evaluate** reclassification using the figworts as an example
6. **Outline** the three domains

# Exit activity: 5 minutes

**On your own and without notes. Hand this in as you leave.**

1. Define clade.
2. State why the molecular clock gives only an estimate.
3. Outline the evidence for the three-domain system.

# Image credits

Mustela erminea upright.jpg – Steve Hillebrand, USFWS – Public domain
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